

Modularity from Architectural Commitment: The Cell as the Modular Unit, Given by Paper 1's Specification Rather Than Evolved

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This work derives from and formalizes the instinct/reasoning separation pattern introduced in *The Instinct/Reasoning Separation Outside the Model* (Li, April 2026), the second paper in the CKS theory series following *Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems* (Li, April 2026). It does not introduce new axioms. Its sole contribution is to articulate, in operational form, a single foundational architectural commitment Paper 2 inherits and elevates: that the Coordination Knowledge Substrate (CKS) is **modular by architectural commitment from the start**, with the cell from Paper 1 as the modular unit, distinguishing CKS modularity from biological modularity, which had to evolve through selection pressure over deep time.

Abstract

Paper 2 builds three architectural levels (cell, aspect, Self), two layers within every cell (DNA and action), an expression mechanism over DNA-layer activation, and a full lifecycle and evolution machinery on top of those primitives. None of that machinery is coherent unless one architectural property is given: that the cell is a *modular unit*, with defined interfaces, composition rules, and interaction surfaces, such that lifecycle and evolution operations can be specified as transformations over modular cells rather than over an undifferentiated whole. Paper 2 names this property by stating that the cell from Paper 1 is the modular unit, like a Lego piece. This note formalizes that statement as a standalone foundational commitment: the modularity is *given* by Paper 1's specification (substrate-cell boundary, composition requirements, three adjacencies, hybrid composition), not evolved as biological modularity was. The note states the commitment, distinguishes it from neighboring notions, names the points at which CKS exceeds biology, identifies the inherited commitments that supply the modularity, and gives an operational test.

1. Why a standalone formalization is needed

Paper 2's three-level structure (B1.02), atomic cell unit (B1.03), aspect (B1.04), Self (B1.05), DNA/action layers (B1.06), and expression mechanism (B1.07) together constitute the structural and intra-cell architecture of a CKS-governed AI Self. These notes describe *what is built*. The next cluster of Paper 2 commitments — birth (B1.09), mating (B1.10), death (B1.11), and the evolution mechanisms (B1.12–B1.16) — describes *how those builds change over time*.

The transition between these clusters depends on a property that has not yet been formalized as standalone: that the cell is a modular unit in a strong architectural sense. Without it, Paper 2's dynamic

operations are not coherently definable. Mating cannot combine two parents into an offspring unless the parents are modular entities with defined boundaries; death cannot retire a cell as an operational unit unless the cell is an operational unit; mutation, directed selection, and action-feedback cannot operate per-cell unless the cell is the locus they operate on. Modularity is the precondition.

Paper 2 names the property in its "Modularity as continuity with Paper 1" subsection, stating that the cell from Paper 1 is the modular unit and that "modularity from the cell up is what makes CKS Selves' evolvability architecturally available rather than evolved." It also names it, contrastively, in its "Modularity from the start rather than evolved" point under "Where CKS exceeds biology." Paper 2 does not separately develop the commitment as standalone foundational architecture. This note does that, closing the structural-and-intra-cell-primitives cluster of Phase B1 (B1.02–B1.08) and preparing the lifecycle and evolution clusters (B1.09–B1.16).

2. The commitment

In the CKS pattern, the architecture is **modular by commitment from the start**, with the cell from Paper 1 as the modular unit. Three properties together constitute the commitment:

1. **The cell is the modular unit.** A CKS artifact in the sense of Paper 1 — substrates plus orchestration rules under human governance — is the unit at which composition and decomposition operate. Paper 2's larger structures (aspects, Selves) and dynamic operations (lifecycle, evolution) take cells as their operands.
2. **The unit has defined modular structure.** Cells have defined boundaries (substrate-cell boundary inherited from A1.02), defined composition rules (composition requirements inherited from A1.13), and defined interaction surfaces (three adjacencies inherited from A1.14). These properties are what make cells composable architecturally.
3. **The modularity is given by Paper 1's specification, not produced by selection.** The architectural pieces (boundary, composition rules, interaction surfaces, hybrid composition pattern from A1.16) are stated commitments of the source pattern. They do not need to evolve into existence under selection pressure; they hold by virtue of the pattern's definition.

Paper 2's framing of the cell as a Lego piece is architectural rather than figurative. A Lego piece is composable because it has standardized studs and tubes (defined interfaces), standardized geometry (composition rules), and a standardized orientation surface for joining (interaction surfaces). The CKS cell is composable on the same terms: the substrate-cell boundary is the standardized interface; the composition requirements specify how cells legally combine; the three adjacencies are the standardized interaction surfaces against which cells join other cells, substrate content, and adjacent components. Paper 1 specifies the piece. Paper 2 specifies what those pieces build into and how those builds develop and change over time. The commitment is foundational rather than derivative: it is given by Paper 1's specification, but Paper 2 elevates it to foundational status because Paper 2's dynamic operations require it.

3. What modularity-from-architectural-commitment is NOT

Five neighboring notions of modularity must be distinguished, each of which would produce a misreading of Paper 2 if substituted.

Not monolithic-with-internal-structure. A model with named submodels or named subroutines inside a single parameter manifold is not modular in the CKS sense. Modularity here requires a defined external boundary against substrate content and adjacent components, not just an internal organization. Such a system does not satisfy A1.02.

Not modular-with-rigid-membership. A system in which cells are bound by intrinsic membership to a single higher structure is not modular in the CKS sense. CKS cells participate in multiple aspects relationally and concurrently per B1.04; their structural roles are purpose-defined, not intrinsic. Rigid membership is a constraint biology operates under because of developmental commitment in differentiated tissues; CKS does not have that constraint.

Not modular-by-emergence. A system whose modularity arose from runtime patterns or post-hoc analysis of internal structure is not modular in the CKS sense. Modularity by architectural commitment is given by the pattern's specification at design time — a stated property of the system's architecture, not an observed property of its behavior.

Not implicit-modularity-that-breaks-under-composition. A system in which a working module of one deployment cannot be composed cleanly with a working module of another — because the modules share hidden state or carry implicit interface assumptions — is not modular in the CKS sense. The composition requirements per A1.13 specify what any multi-substrate composition must satisfy.

Not evolved-modularity. A system whose modular structure was produced by selection pressure under evolvability advantages is biologically modular but not modular in the CKS architectural sense. Biology selected for modularity because modular systems evolve faster; the modularity took deep time and produced compromises along the way. CKS gets modularity as a stated property of its specification.

The commitment also has four limits. Modularity does *not* mean cells are immutable (they evolve through Paper 2's mechanisms and pass through lifecycle); does *not* mean cells are isolated (they compose through A1.13 and interact across A1.14); does *not* mean cells are single-purpose (they participate in multiple aspects per B1.04); and does *not* eliminate intra-cell complexity (cells have two internal layers per B1.06 and a governed expression mechanism per B1.07). What is given is the modular structure of the unit, not the contents the unit holds.

4. The biological analog and where CKS exceeds biology

The biological analog is modularity at the cell-tissue-organism levels. Biological cells are modular units with defined boundaries (membranes), defined composition into tissues (developmental rules), and standardized interaction surfaces (receptors, junctions, signaling pathways). Tissues compose into organs; organs into organisms. Paper 2's three architectural levels are parallel.

The decisive difference is *origin*. Biology evolved this modularity over hundreds of millions of years because evolutionary pressures select for evolvability, and modular systems evolve faster than non-modular ones. The selection pressure produced modularity, but slowly, and with compromises that

follow from modularity-as-evolved-property:

- **Rigid tissue and organ memberships.** Biological cells are committed by developmental fate to a specific tissue context; reassignment is not architecturally available. CKS cells participate in multiple aspects simultaneously per B1.04 — modular cells with relational rather than intrinsic membership.
- **Species barriers.** Biological lineages cannot mate across species boundaries because the inheritance channels are not compatible. CKS supports cross-lineage mating per B1.10 because the modularity is architectural and orchestration rules can govern the combination.
- **Separate inheritance channels.** Biology has genetic, epigenetic, and cultural inheritance as separate channels because they evolved separately and at different timescales. CKS collapses them into one mating primitive over content combination because the modularity is uniform across DNA and action layers.
- **Slow body-plan evolution.** Biological body plans take millions of years to evolve because the evolvability of large structural change is itself constrained by what modular architectures evolution has already produced. CKS Selves restructure on operational timescales through governed reorganization (vertical evolution per B1.16) — reorganization is a substrate operation, not a developmental rewrite.
- **No reversibility.** Extinct biological species cannot return because the modular structure of their cells is not preserved anywhere addressable. Archived CKS cells remain addressable per Paper 2 §6.4 and can be reactivated if purpose changes, because modularity is preserved in substrate as authoritative state per A1.08.

The point is not that CKS is "better than biology" in some general sense. Biology operates under constraints that follow from modularity-as-evolved-property; CKS, taking modularity as architectural commitment from the start, does not inherit those constraints. The advantages above are direct consequences of where CKS does not need the workarounds biology developed.

5. The inherited Paper 1 commitments that make modularity given

Modularity-from-architectural-commitment is given by four Paper 1 commitments composed together.

A1.02 — The substrate-cell boundary is the cell's external interface, distinguishing substrate content from cell behavior. It is the standardized interface that makes cells composable.

A1.13 — Composition requirements specifies the rules under which cells legally combine: shared content addressability, conflict-handling discipline across composed cells, governance authority over combined orchestration rules, retraceability across composition seams, and cost-shape preservation.

A1.14 — Three adjacencies specifies the architectural neighbors against which the modular cell is exercised: parametric memory of the LLM, retrieval-augmented generation against external corpora, and external structured memory. The three adjacencies are the standardized interaction surfaces.

A1.16 — Hybrid systems composition specifies the three patterns under which CKS substrates compose with adjacent AI components — input pattern, derived view pattern, and separate concern

pattern — extending modularity from cell-to-cell to cell-to-non-cell composition.

These four commitments together supply the modular structure: the substrate-cell boundary is the interface; the composition requirements are the combination rules; the three adjacencies are the interaction surfaces; the hybrid composition is the pattern for non-cell counterparts. Modularity is not a fifth commitment added on top; it is the property the four commitments confer on the cell unit, and it is what Paper 2 elevates to explicit foundational status because Paper 2's machinery requires it.

6. Operational implications

Six consequences follow directly for Paper 2's machinery. *Deployments organize around cells as modular units* — cells are the units of organization, of provenance, and of substrate addressing. *Lifecycle operations operate on cells as modular entities*: birth (B1.09), mating (B1.10), and death (B1.11) take modular cells as operands. *Evolution mechanisms operate on modular cells*: instinct evolution (B1.12), DNA evolution (B1.13), action-feedback evolution (B1.14), and multi-level evolution (B1.15) all preserve modularity across the per-cell, per-aspect, and per-Self loci on which they operate. *Aspects and Selves compose modular cells*: the same modular cell can participate in multiple aspects relationally per B1.04, and a Self holds heterogeneous aspects coherently because modularity is uniform at the cell level. *Modular cells are vendor-portable*: combined with tool-agnosticism per A1.05, a cell's substrate-cell boundary, composition requirements, and three adjacencies are the portable architectural specification, while vendors and runtimes are below this specification. *Modularity is testable*: cell-boundary tests check A1.02, composition-requirements tests check A1.13, three-adjacencies tests check A1.14, hybrid-composition tests check A1.16.

7. Operational test

A Paper 2 instantiation satisfies the modularity-from-architectural-commitment foundational commitment if and only if all of the following hold:

1. The atomic unit of organization is a CKS cell in the sense of Paper 1, with substrates, orchestration rules, and the architectural commitments Paper 1 defends.
2. The cell's external interface is the substrate-cell boundary per A1.02, defined and respected as an architectural property of the design rather than as a runtime convention.
3. Combinations of cells satisfy the composition requirements per A1.13.
4. The cell's interactions with adjacent components are organized by the three adjacencies per A1.14.
5. The cell's composition with non-CKS adjacent AI components follows one of the hybrid composition patterns per A1.16.
6. The modularity is given by the architectural specification — stated, not observed; designed, not selected — and Paper 2's lifecycle and evolution operations take the cell as their operand on the strength of (1)–(5).

A system that fails any of (1)–(6) may be modular in some other sense but is not modular by architectural commitment in the CKS sense. The failure mode that matters most is (6): a system that

exhibits modular behavior at runtime but whose modularity is not an architectural property of its specification cannot guarantee that lifecycle and evolution operations will preserve modularity across changes.

8. Conclusion

CKS is modular by architectural commitment from the start. The cell from Paper 1 is the modular unit — a Lego piece in the architectural sense, with the substrate-cell boundary as defined interface, the composition requirements as combination rules, and the three adjacencies as standardized interaction surfaces. Paper 1 specifies the piece; Paper 2 specifies what those pieces build into and how those builds develop and change over time.

Biology had to evolve modular architectures over deep time because modularity supports evolvability and selection pressure favors evolvable systems. The compromise is rigid tissue and organ memberships, species barriers, separate inheritance channels, slow body-plan evolution, and no reversibility. CKS, taking modularity as architectural commitment from the start, does not inherit those constraints and gains the corresponding architectural advantages: relational role membership, cross-lineage mating compatibility, single mating primitive over content combination, structural evolvability on operational timescales, and addressable archival of retired cells.

This note closes the structural-and-intra-cell-primitives cluster of Phase B1 (B1.02–B1.08), which establishes *what is built* in Paper 2's static architecture. The next clusters — lifecycle (B1.09–B1.11), evolution mechanisms (B1.12–B1.15), bidirectional evolution (B1.16), and structural properties (B1.17–B1.20) — establish *how those builds develop and change over time*. The bridge between them is modularity by architectural commitment: the property that makes the static architecture's operands available to the dynamic machinery as defined units. Subsequent work that depends on modularity-by-emergence, modularity-by-runtime-behavior, or modularity-by-selection is using a different architectural commitment, and the difference should be named.

Source papers

Li, W. (2026). *Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems*. April 2026. ORCID: 0009-0004-8065-3235.

Li, W. (2026). *The Instinct/Reasoning Separation Outside the Model: Extending the Coordination Knowledge Substrate Pattern to AI Selves under Human Governance*. April 2026. ORCID: 0009-0004-8065-3235.

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